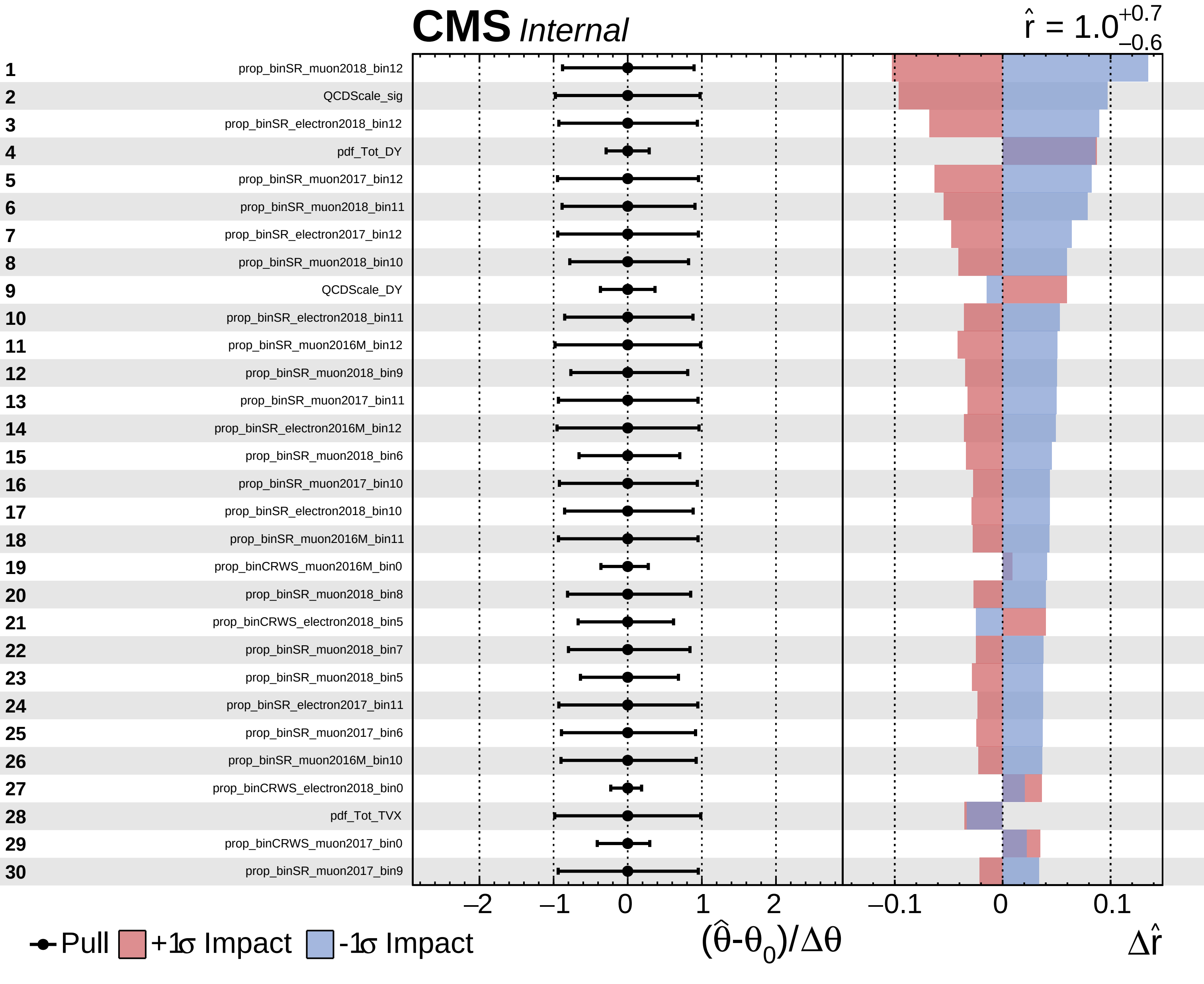
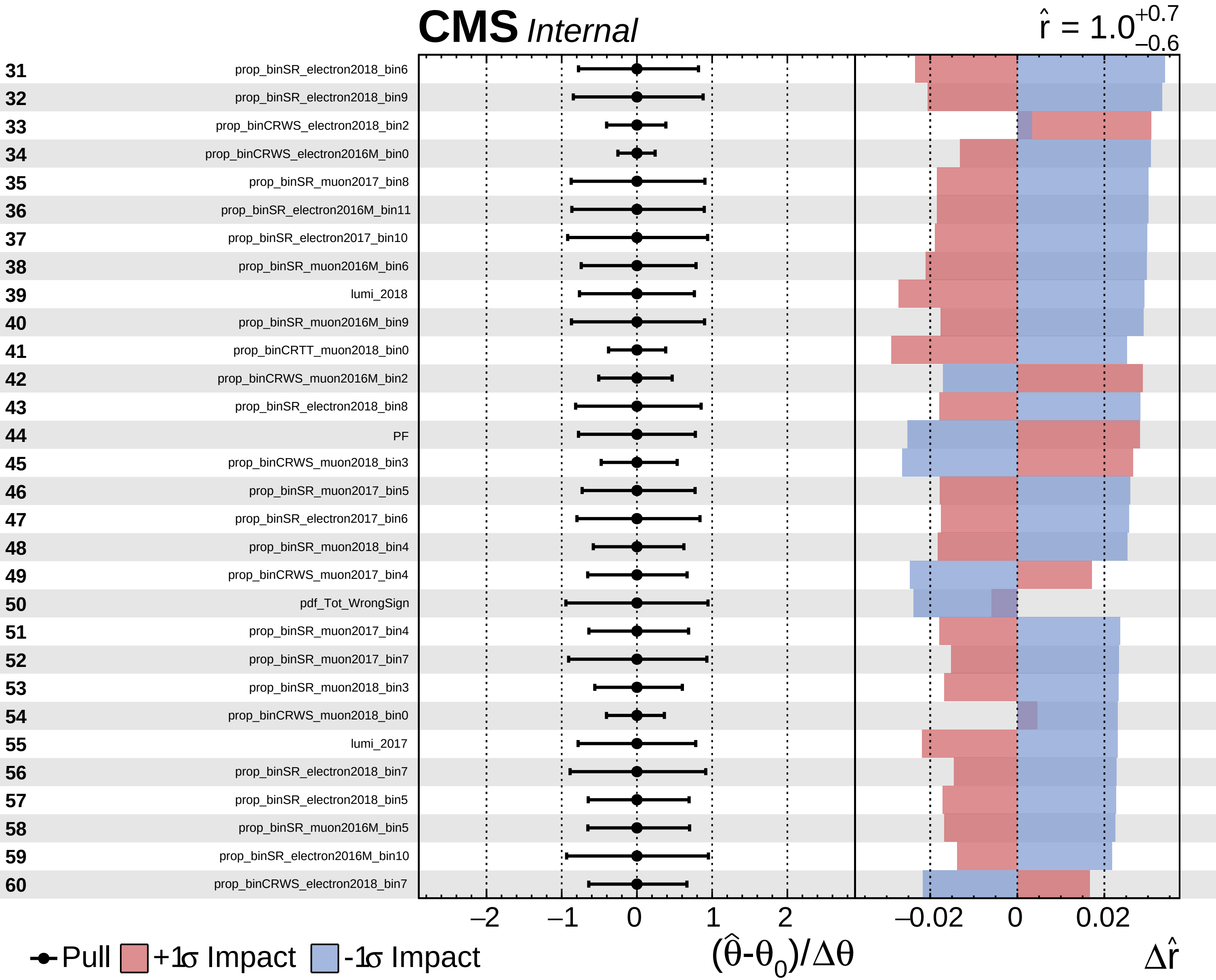
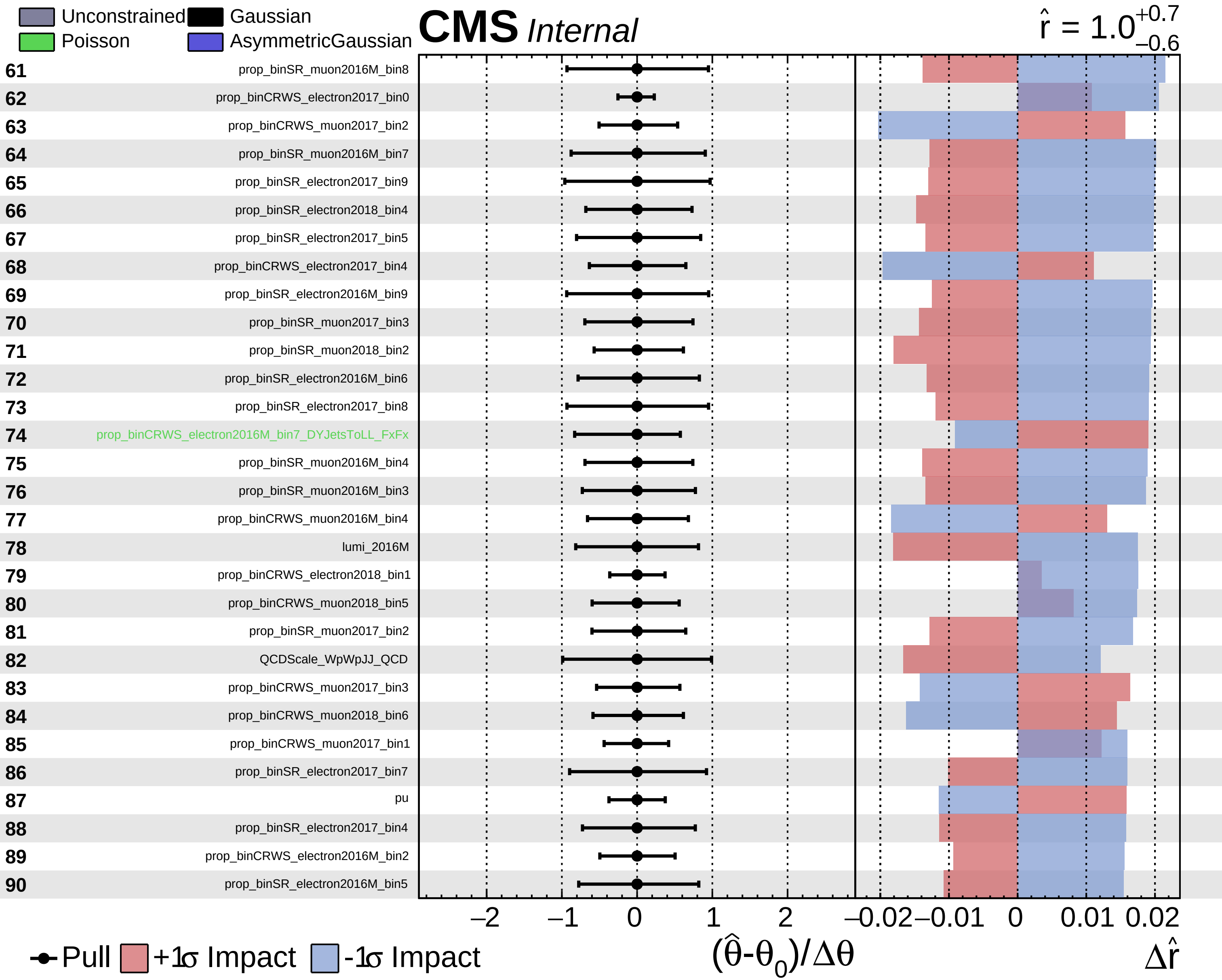
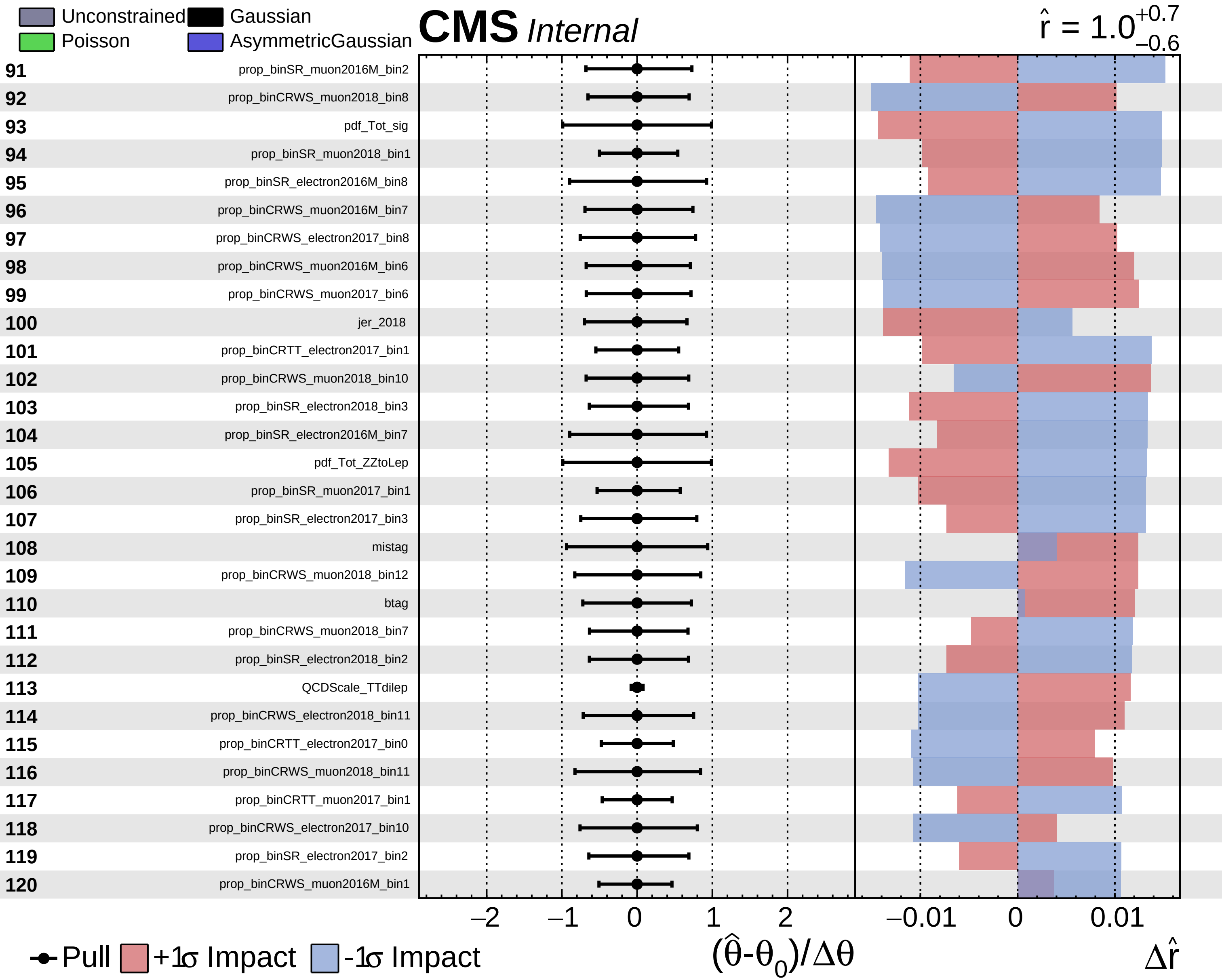

CMS Internal
r̂ = 1.0 +0.7 −0.6
1 prop_binSR_muon2018_bin12
2 QCDScale_sig
3 prop_binSR_electron2018_bin12
4 pdf_Tot_DY
5 prop_binSR_muon2017_bin12
6 prop_binSR_muon2018_bin11
7 prop_binSR_electron2017_bin12
8 prop_binSR_muon2018_bin10
9 QCDScale_DY
10 prop_binSR_electron2018_bin11
11 prop_binSR_muon2016M_bin12
12 prop_binSR_muon2018_bin9
13 prop_binSR_muon2017_bin11
14 prop_binSR_electron2016M_bin12
15 prop_binSR_muon2018_bin6
16 prop_binSR_muon2017_bin10
17 prop_binSR_electron2018_bin10
18 prop_binSR_muon2016M_bin11
19 prop_binCRWS_muon2016M_bin0
20 prop_binSR_muon2018_bin8
21 prop_binCRWS_electron2018_bin5
22 prop_binSR_muon2018_bin7
23 prop_binSR_muon2018_bin5
24 prop_binSR_electron2017_bin11
25 prop_binSR_muon2017_bin6
26 prop_binSR_muon2016M_bin10
27 prop_binCRWS_electron2018_bin0
28 pdf_Tot_TVX
29 prop_binCRWS_muon2017_bin0
30 prop_binSR_muon2017_bin9
−2 −1 0 1 2
−0.1 0 0.1
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δr̂

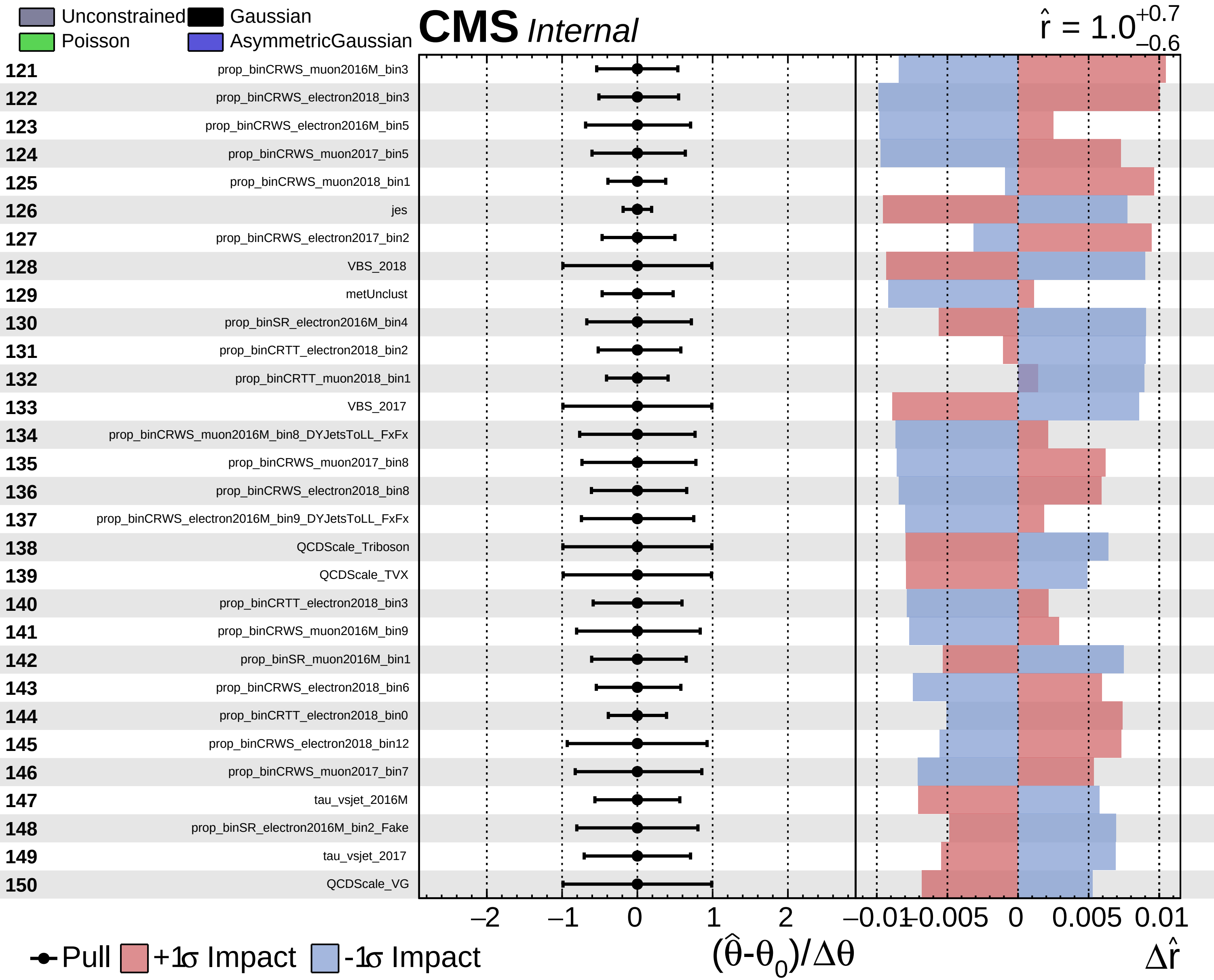

CMS Internal
r̂ = 1.0 +0.7 −0.6
31 prop_binSR_electron2018_bin6
32 prop_binSR_electron2018_bin9
33 prop_binCRWS_electron2018_bin2
34 prop_binCRWS_electron2016M_bin0
35 prop_binSR_muon2017_bin8
36 prop_binSR_electron2016M_bin11
37 prop_binSR_electron2017_bin10
38 prop_binSR_muon2016M_bin6
39 lumi_2018
40 prop_binSR_muon2016M_bin9
41 prop_binCRTT_muon2018_bin0
42 prop_binCRWS_muon2016M_bin2
43 prop_binSR_electron2018_bin8
44 PF
45 prop_binCRWS_muon2018_bin3
46 prop_binSR_muon2017_bin5
47 prop_binSR_electron2017_bin6
48 prop_binSR_muon2018_bin4
49 prop_binCRWS_muon2017_bin4
50 pdf_Tot_WrongSign
51 prop_binSR_muon2017_bin4
52 prop_binSR_muon2017_bin7
53 prop_binSR_muon2018_bin3
54 prop_binCRWS_muon2018_bin0
55 lumi_2017
56 prop_binSR_electron2018_bin7
57 prop_binSR_electron2018_bin5
58 prop_binSR_muon2016M_bin5
59 prop_binSR_electron2016M_bin10
60 prop_binCRWS_electron2018_bin7
−2
−1
0
1
2
−0.02
0
0.02
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂



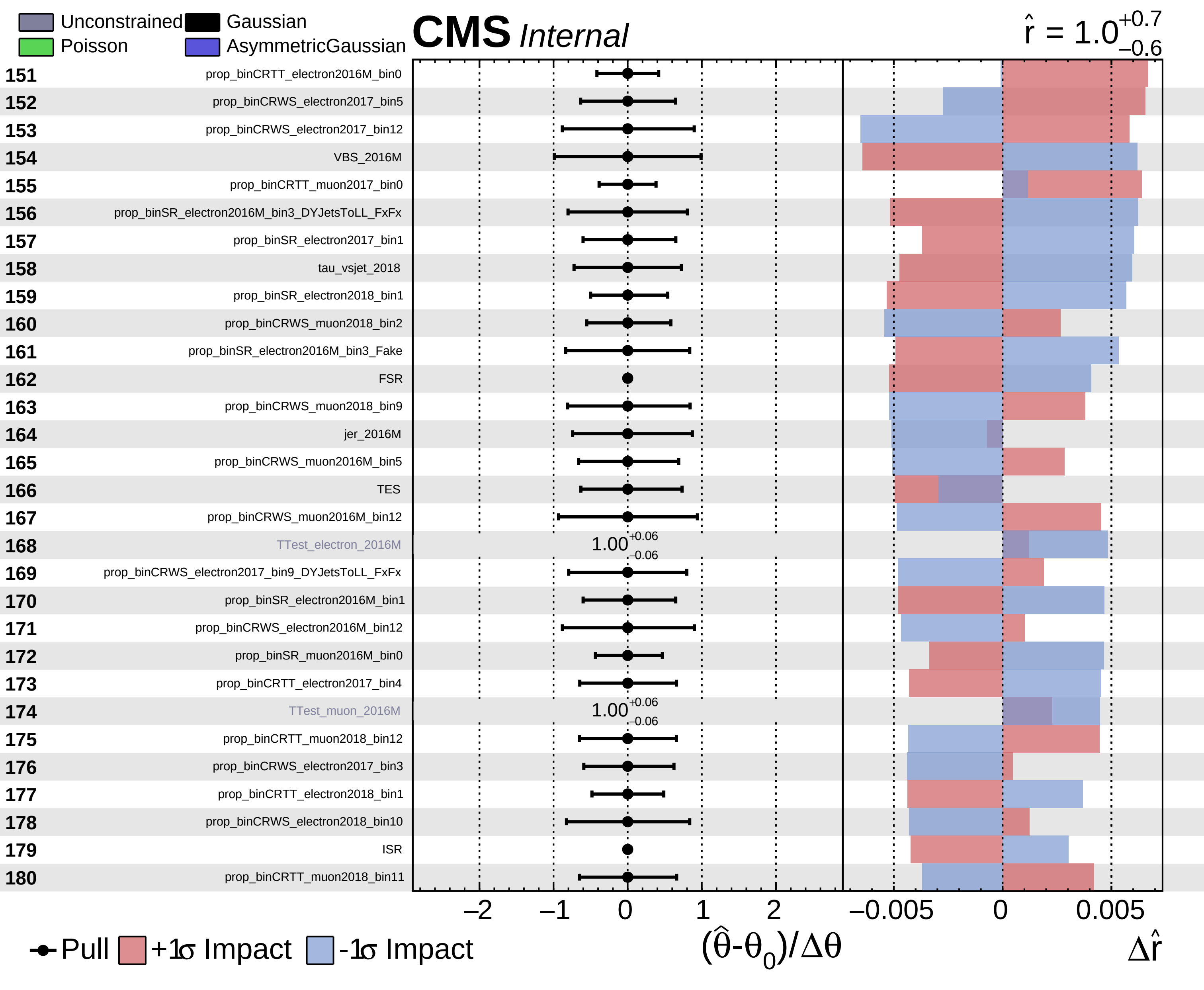
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.7}_{-0.6}$
61 prop_binSR_muon2016M_bin8
62 prop_binCRWS_electron2017_bin0
63 prop_binCRWS_muon2017_bin2
64 prop_binSR_muon2016M_bin7
65 prop_binSR_electron2017_bin9
66 prop_binSR_electron2018_bin4
67 prop_binSR_electron2017_bin5
68 prop_binCRWS_electron2017_bin4
69 prop_binSR_electron2016M_bin9
70 prop_binSR_muon2017_bin3
71 prop_binSR_muon2018_bin2
72 prop_binSR_electron2016M_bin6
73 prop_binSR_electron2017_bin8
74 prop_binCRWS_electron2016M_bin7_DYJetsToLL_FxFx
75 prop_binSR_muon2016M_bin4
76 prop_binSR_muon2016M_bin3
77 prop_binCRWS_muon2016M_bin4
78 lumi_2016M
79 prop_binCRWS_electron2018_bin1
80 prop_binCRWS_muon2018_bin5
81 prop_binSR_muon2017_bin2
82 QCDScale_WpWpJJ_QCD
83 prop_binCRWS_muon2017_bin3
84 prop_binCRWS_muon2018_bin6
85 prop_binCRWS_muon2017_bin1
86 prop_binSR_electron2017_bin7
87 pu
88 prop_binSR_electron2017_bin4
89 prop_binCRWS_electron2016M_bin2
90 prop_binSR_electron2016M_bin5
−2 −1 0 1 2
−0.02 −0.01 0 0.01 0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



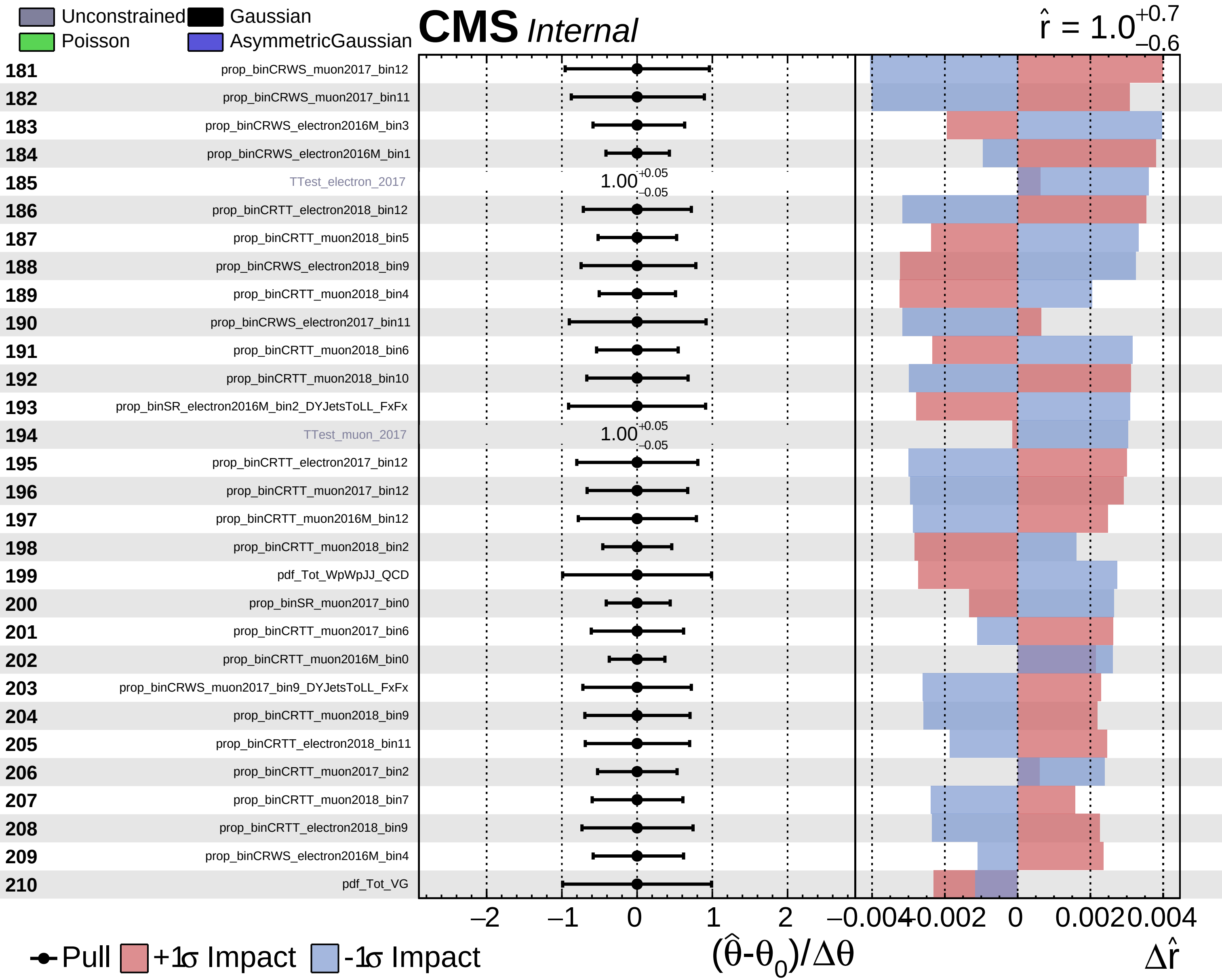
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.7}_{-0.6}$
91 prop_binSR_muon2016M_bin2
92 prop_binCRWS_muon2018_bin8
93 pdf_Tot_sig
94 prop_binSR_muon2018_bin1
95 prop_binSR_electron2016M_bin8
96 prop_binCRWS_muon2016M_bin7
97 prop_binCRWS_electron2017_bin8
98 prop_binCRWS_muon2016M_bin6
99 prop_binCRWS_muon2017_bin6
100 jer_2018
101 prop_binCRTT_electron2017_bin1
102 prop_binCRWS_muon2018_bin10
103 prop_binSR_electron2018_bin3
104 prop_binSR_electron2016M_bin7
105 pdf_Tot_ZZtoLep
106 prop_binSR_muon2017_bin1
107 prop_binSR_electron2017_bin3
108 mistag
109 prop_binCRWS_muon2018_bin12
110 btag
111 prop_binCRWS_muon2018_bin7
112 prop_binSR_electron2018_bin2
113 QCDScale_TTdilep
114 prop_binCRWS_electron2018_bin11
115 prop_binCRTT_electron2017_bin0
116 prop_binCRWS_muon2018_bin11
117 prop_binCRTT_muon2017_bin1
118 prop_binCRWS_electron2017_bin10
119 prop_binSR_electron2017_bin2
120 prop_binCRWS_muon2016M_bin1
−2 −1 0 1 2
−0.01 0 0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



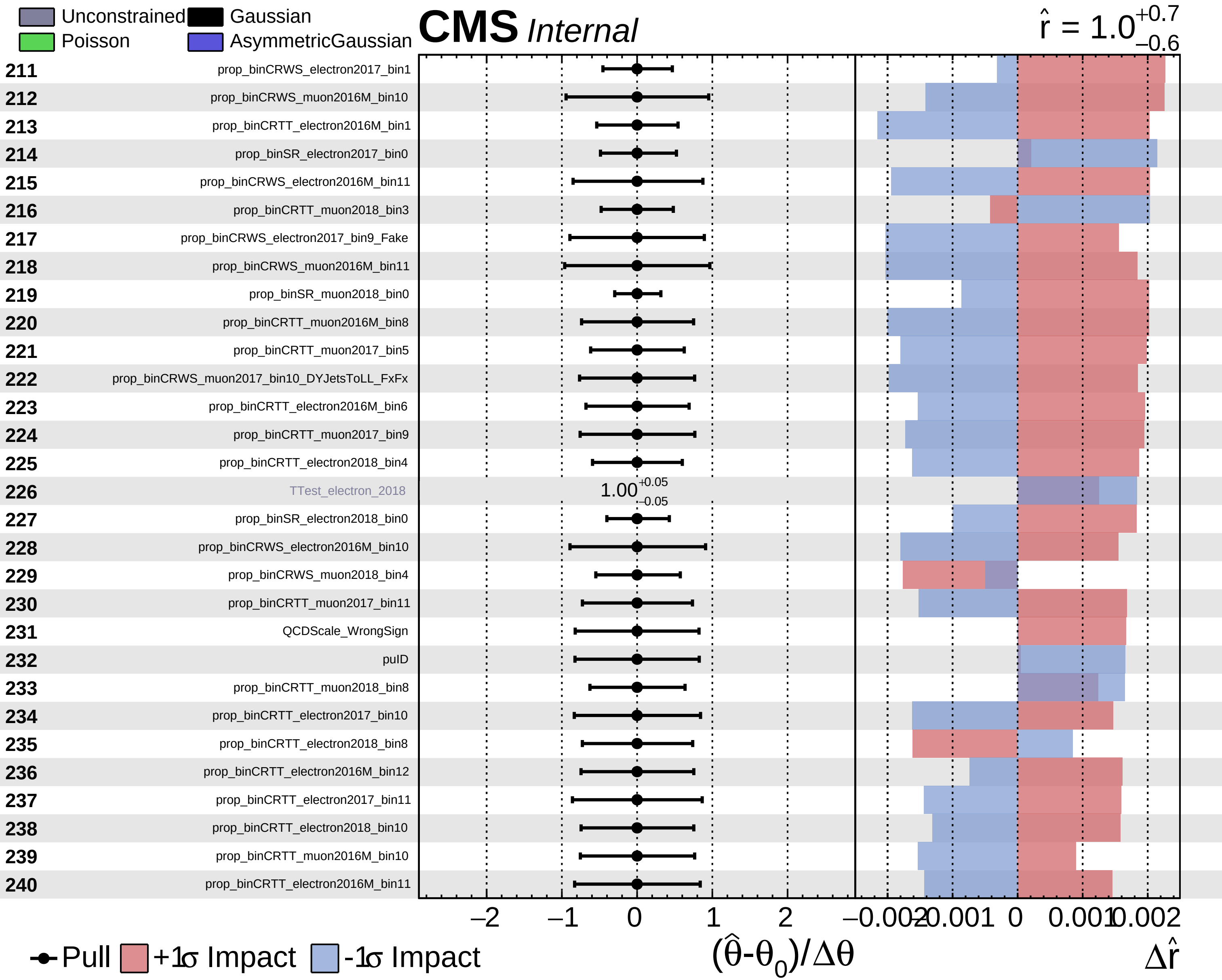
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.7}_{-0.6}$
121 prop_binCRWS_muon2016M_bin3
122 prop_binCRWS_electron2018_bin3
123 prop_binCRWS_electron2016M_bin5
124 prop_binCRWS_muon2017_bin5
125 prop_binCRWS_muon2018_bin1
126 jes
127 prop_binCRWS_electron2017_bin2
128 VBS_2018
129 metUnclust
130 prop_binSR_electron2016M_bin4
131 prop_binCRTT_electron2018_bin2
132 prop_binCRTT_muon2018_bin1
133 VBS_2017
134 prop_binCRWS_muon2016M_bin8_DYJetsToLL_FxFx
135 prop_binCRWS_muon2017_bin8
136 prop_binCRWS_electron2018_bin8
137 prop_binCRWS_electron2016M_bin9_DYJetsToLL_FxFx
138 QCDScale_Triboson
139 QCDScale_TVX
140 prop_binCRTT_electron2018_bin3
141 prop_binCRWS_muon2016M_bin9
142 prop_binSR_muon2016M_bin1
143 prop_binCRWS_electron2018_bin6
144 prop_binCRTT_electron2018_bin0
145 prop_binCRWS_electron2018_bin12
146 prop_binCRWS_muon2017_bin7
147 tau_vsjet_2016M
148 prop_binSR_electron2016M_bin2_Fake
149 tau_vsjet_2017
150 QCDScale_VG
−2 −1 0 1 2
−0.01 −0.005 0 0.005 0.01
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

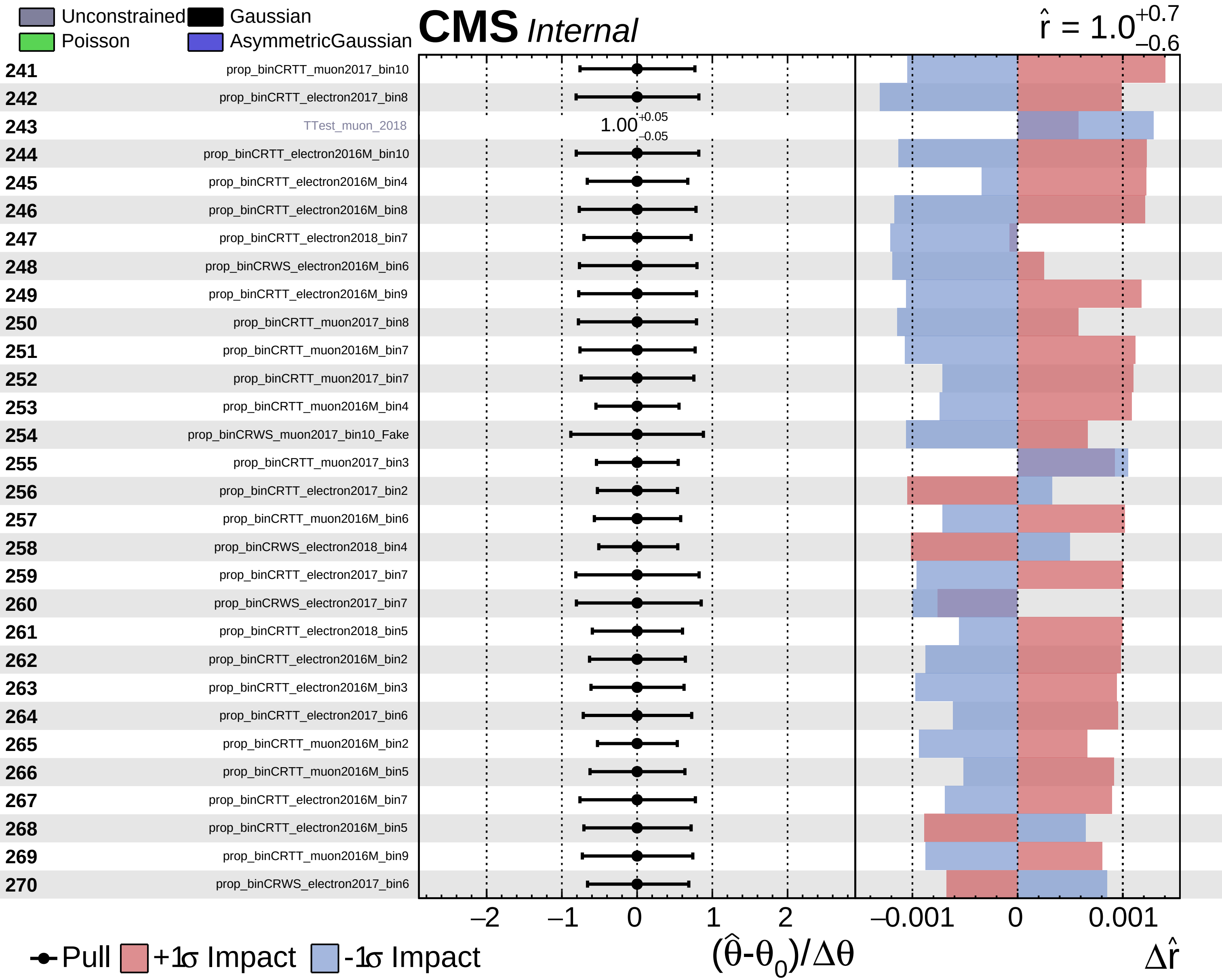


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.7}_{-0.6}$
151 prop_binCRTT_electron2016M_bin0
152 prop_binCRWS_electron2017_bin5
153 prop_binCRWS_electron2017_bin12
154 VBS_2016M
155 prop_binCRTT_muon2017_bin0
156 prop_binSR_electron2016M_bin3_DYJetsToLL_FxFx
157 prop_binSR_electron2017_bin1
158 tau_vsjet_2018
159 prop_binSR_electron2018_bin1
160 prop_binCRWS_muon2018_bin2
161 prop_binSR_electron2016M_bin3_Fake
162 FSR
163 prop_binCRWS_muon2018_bin9
164 jer_2016M
165 prop_binCRWS_muon2016M_bin5
166 TES
167 prop_binCRWS_muon2016M_bin12
168 TTest_electron_2016M
$1.00^{+0.06}_{-0.06}$
169 prop_binCRWS_electron2017_bin9_DYJetsToLL_FxFx
170 prop_binSR_electron2016M_bin1
171 prop_binCRWS_electron2016M_bin12
172 prop_binSR_muon2016M_bin0
173 prop_binCRTT_electron2017_bin4
174 TTest_muon_2016M
$1.00^{+0.06}_{-0.06}$
175 prop_binCRTT_muon2018_bin12
176 prop_binCRWS_electron2017_bin3
177 prop_binCRTT_electron2018_bin1
178 prop_binCRWS_electron2018_bin10
179 ISR
180 prop_binCRTT_muon2018_bin11
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

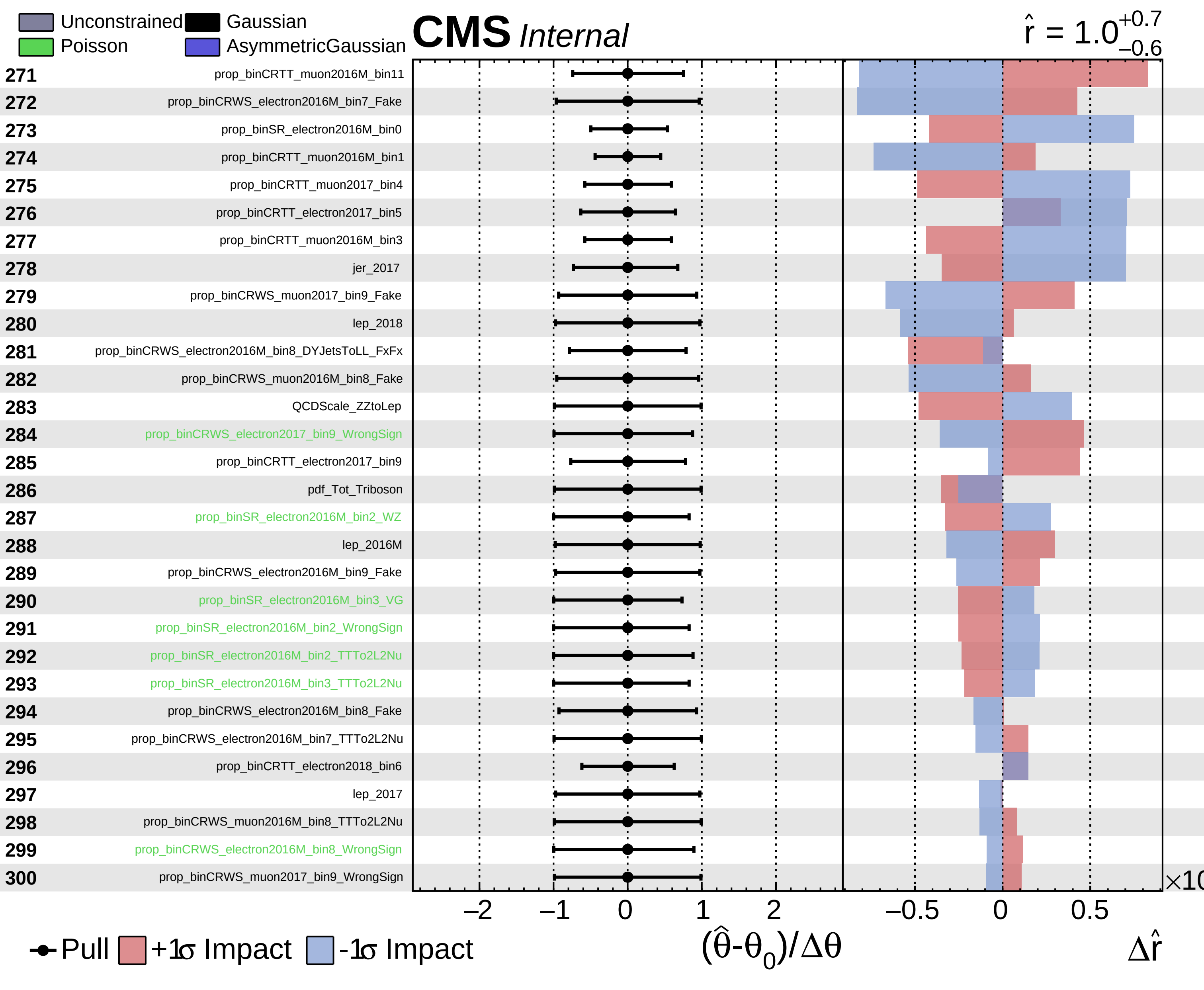


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.7}_{-0.6}$
181 prop_binCRWS_muon2017_bin12
182 prop_binCRWS_muon2017_bin11
183 prop_binCRWS_electron2016M_bin3
184 prop_binCRWS_electron2016M_bin1
185 TTest_electron_2017 $1.00^{+0.05}_{-0.05}$
186 prop_binCRTT_electron2018_bin12
187 prop_binCRTT_muon2018_bin5
188 prop_binCRWS_electron2018_bin9
189 prop_binCRTT_muon2018_bin4
190 prop_binCRWS_electron2017_bin11
191 prop_binCRTT_muon2018_bin6
192 prop_binCRTT_muon2018_bin10
193 prop_binSR_electron2016M_bin2_DYJetsToLL_FxFx
194 TTest_muon_2017 $1.00^{+0.05}_{-0.05}$
195 prop_binCRTT_electron2017_bin12
196 prop_binCRTT_muon2017_bin12
197 prop_binCRTT_muon2016M_bin12
198 prop_binCRTT_muon2018_bin2
199 pdf_Tot_WpWpJJ_QCD
200 prop_binSR_muon2017_bin0
201 prop_binCRTT_muon2017_bin6
202 prop_binCRTT_muon2016M_bin0
203 prop_binCRWS_muon2017_bin9_DYJetsToLL_FxFx
204 prop_binCRTT_muon2018_bin9
205 prop_binCRTT_electron2018_bin11
206 prop_binCRTT_muon2017_bin2
207 prop_binCRTT_muon2018_bin7
208 prop_binCRTT_electron2018_bin9
209 prop_binCRWS_electron2016M_bin4
210 pdf_Tot_VG
−2 −1 0 1 2
−0.004 −0.002 0 0.002 0.004
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

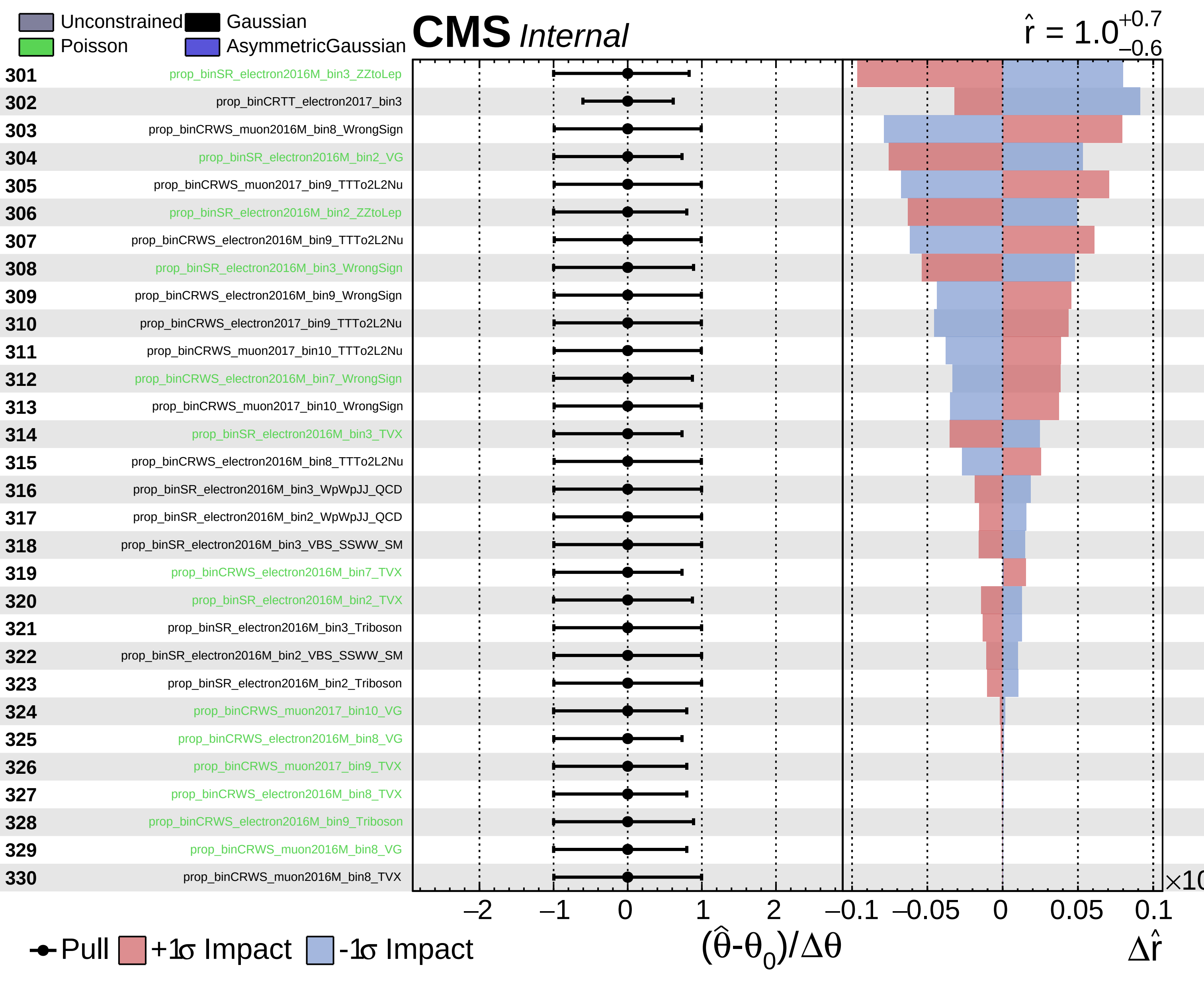

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.7}_{-0.6}$
211 prop_binCRWS_electron2017_bin1
212 prop_binCRWS_muon2016M_bin10
213 prop_binCRTT_electron2016M_bin1
214 prop_binSR_electron2017_bin0
215 prop_binCRWS_electron2016M_bin11
216 prop_binCRTT_muon2018_bin3
217 prop_binCRWS_electron2017_bin9_Fake
218 prop_binCRWS_muon2016M_bin11
219 prop_binSR_muon2018_bin0
220 prop_binCRTT_muon2016M_bin8
221 prop_binCRTT_muon2017_bin5
222 prop_binCRWS_muon2017_bin10_DYJetsToLL_FxFx
223 prop_binCRTT_electron2016M_bin6
224 prop_binCRTT_muon2017_bin9
225 prop_binCRTT_electron2018_bin4
226 TTest_electron_2018
$1.00^{+0.05}_{-0.05}$
227 prop_binSR_electron2018_bin0
228 prop_binCRWS_electron2016M_bin10
229 prop_binCRWS_muon2018_bin4
230 prop_binCRTT_muon2017_bin11
231 QCDScale_WrongSign
232 puID
233 prop_binCRTT_muon2018_bin8
234 prop_binCRTT_electron2017_bin10
235 prop_binCRTT_electron2018_bin8
236 prop_binCRTT_electron2016M_bin12
237 prop_binCRTT_electron2017_bin11
238 prop_binCRTT_electron2018_bin10
239 prop_binCRTT_muon2016M_bin10
240 prop_binCRTT_electron2016M_bin11
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



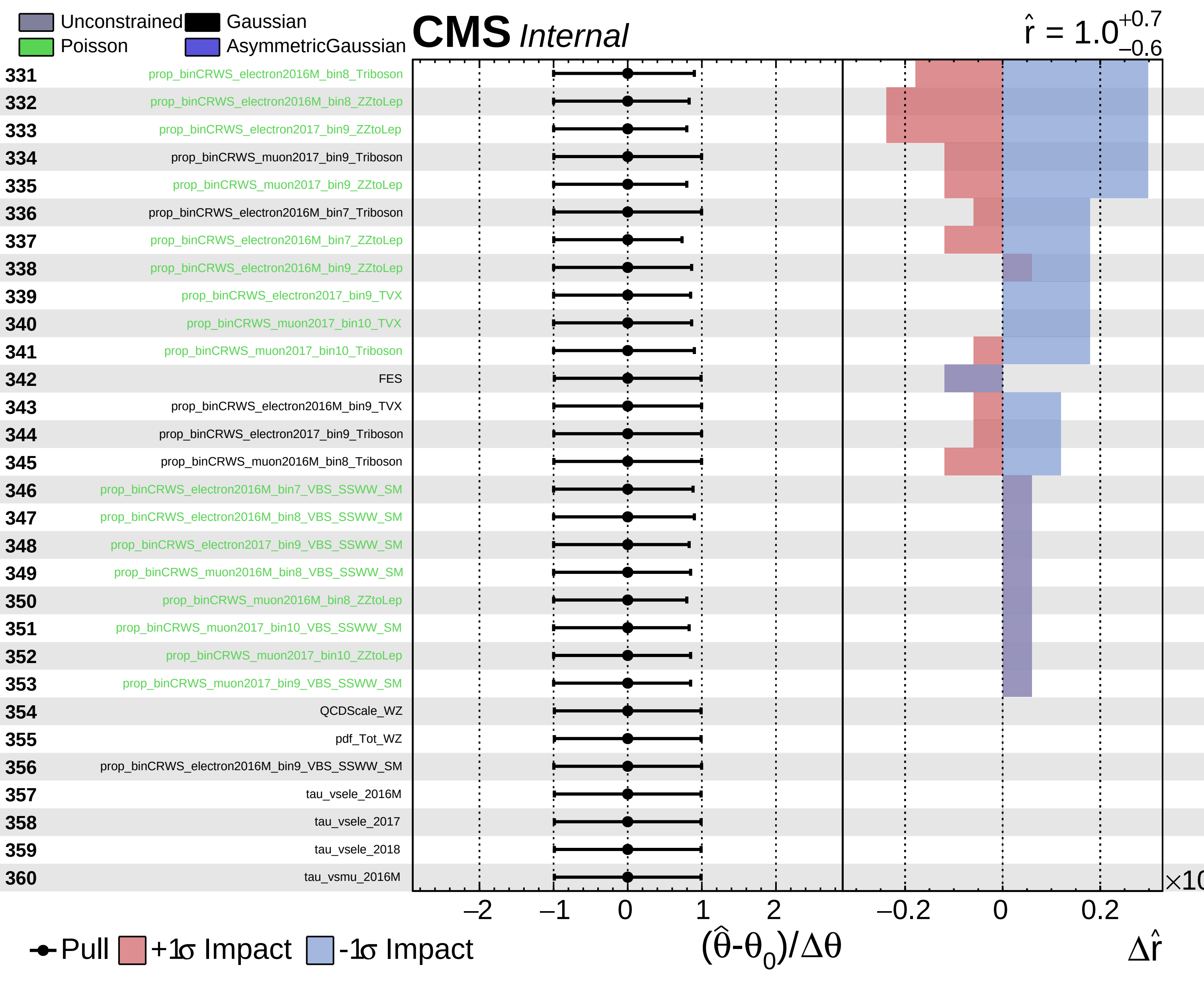
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.7}_{-0.6}$
241 prop_binCRTT_muon2017_bin10
242 prop_binCRTT_electron2017_bin8
243 TTest_muon_2018
$1.00^{+0.05}_{-0.05}$
244 prop_binCRTT_electron2016M_bin10
245 prop_binCRTT_electron2016M_bin4
246 prop_binCRTT_electron2016M_bin8
247 prop_binCRTT_electron2018_bin7
248 prop_binCRWS_electron2016M_bin6
249 prop_binCRTT_electron2016M_bin9
250 prop_binCRTT_muon2017_bin8
251 prop_binCRTT_muon2016M_bin7
252 prop_binCRTT_muon2017_bin7
253 prop_binCRTT_muon2016M_bin4
254 prop_binCRWS_muon2017_bin10_Fake
255 prop_binCRTT_muon2017_bin3
256 prop_binCRTT_electron2017_bin2
257 prop_binCRTT_muon2016M_bin6
258 prop_binCRWS_electron2018_bin4
259 prop_binCRTT_electron2017_bin7
260 prop_binCRWS_electron2017_bin7
261 prop_binCRTT_electron2018_bin5
262 prop_binCRTT_electron2016M_bin2
263 prop_binCRTT_electron2016M_bin3
264 prop_binCRTT_electron2017_bin6
265 prop_binCRTT_muon2016M_bin2
266 prop_binCRTT_muon2016M_bin5
267 prop_binCRTT_electron2016M_bin7
268 prop_binCRTT_electron2016M_bin5
269 prop_binCRTT_muon2016M_bin9
270 prop_binCRWS_electron2017_bin6
−2 −1 0 1 2
−0.001 0 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



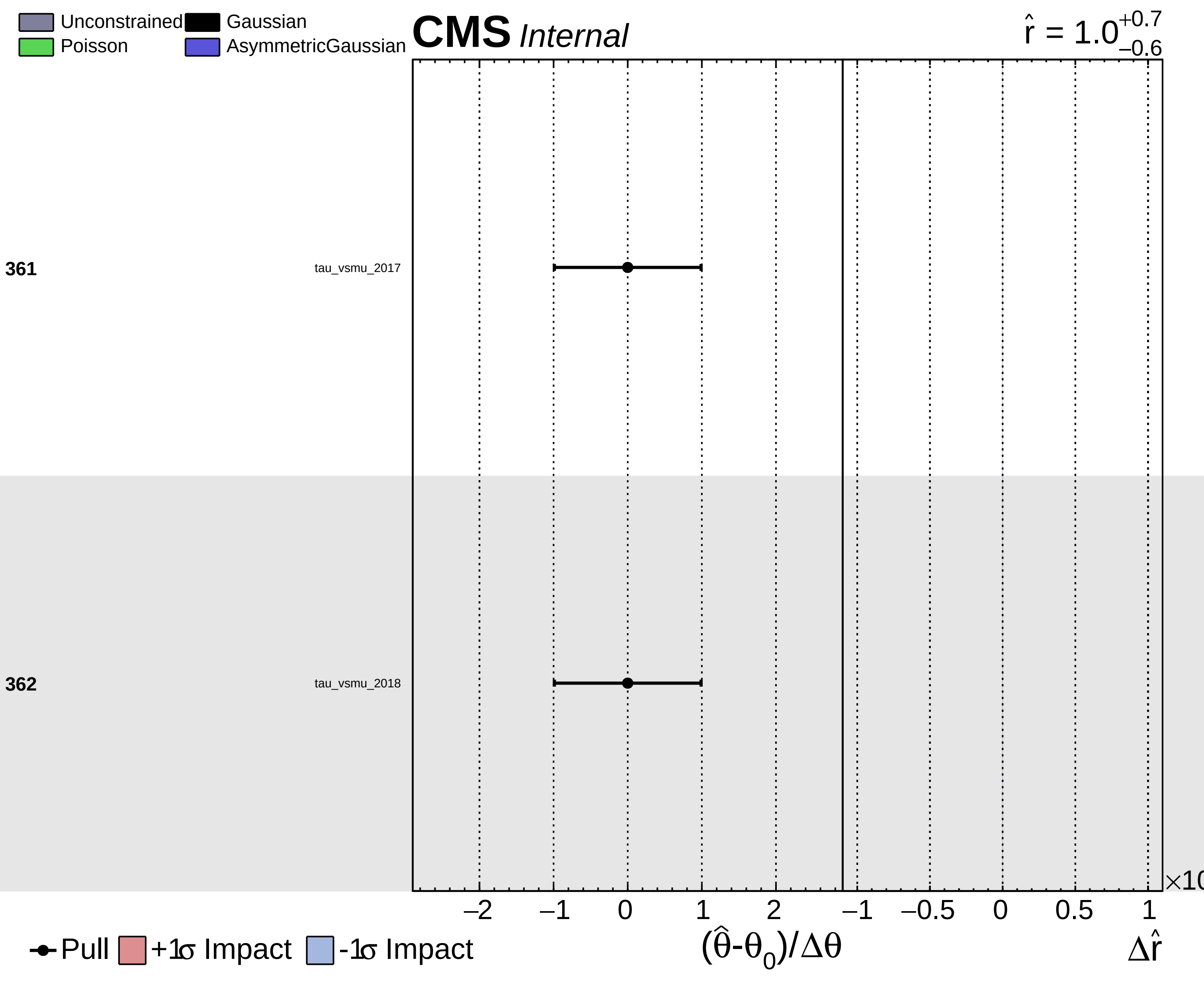
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.7}_{-0.6}$
271 prop_binCRTT_muon2016M_bin11
272 prop_binCRWS_electron2016M_bin7_Fake
273 prop_binSR_electron2016M_bin0
274 prop_binCRTT_muon2016M_bin1
275 prop_binCRTT_muon2017_bin4
276 prop_binCRTT_electron2017_bin5
277 prop_binCRTT_muon2016M_bin3
278 jer_2017
279 prop_binCRWS_muon2017_bin9_Fake
280 lep_2018
281 prop_binCRWS_electron2016M_bin8_DYJetsToLL_FxFx
282 prop_binCRWS_muon2016M_bin8_Fake
283 QCDScale_ZZtoLep
284 prop_binCRWS_electron2017_bin9_WrongSign
285 prop_binCRTT_electron2017_bin9
286 pdf_Tot_Triboson
287 prop_binSR_electron2016M_bin2_WZ
288 lep_2016M
289 prop_binCRWS_electron2016M_bin9_Fake
290 prop_binSR_electron2016M_bin3_VG
291 prop_binSR_electron2016M_bin2_WrongSign
292 prop_binSR_electron2016M_bin2_TTTo2L2Nu
293 prop_binSR_electron2016M_bin3_TTTo2L2Nu
294 prop_binCRWS_electron2016M_bin8_Fake
295 prop_binCRWS_electron2016M_bin7_TTTo2L2Nu
296 prop_binCRTT_electron2018_bin6
297 lep_2017
298 prop_binCRWS_muon2016M_bin8_TTTo2L2Nu
299 prop_binCRWS_electron2016M_bin8_WrongSign
300 prop_binCRWS_muon2017_bin9_WrongSign
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.7}_{-0.6}$
301 prop_binSR_electron2016M_bin3_ZZtoLep
302 prop_binCRTT_electron2017_bin3
303 prop_binCRWS_muon2016M_bin8_WrongSign
304 prop_binSR_electron2016M_bin2_VG
305 prop_binCRWS_muon2017_bin9_TTTo2L2Nu
306 prop_binSR_electron2016M_bin2_ZZtoLep
307 prop_binCRWS_electron2016M_bin9_TTTo2L2Nu
308 prop_binSR_electron2016M_bin3_WrongSign
309 prop_binCRWS_electron2016M_bin9_WrongSign
310 prop_binCRWS_electron2017_bin9_TTTo2L2Nu
311 prop_binCRWS_muon2017_bin10_TTTo2L2Nu
312 prop_binCRWS_electron2016M_bin7_WrongSign
313 prop_binCRWS_muon2017_bin10_WrongSign
314 prop_binSR_electron2016M_bin3_TVX
315 prop_binCRWS_electron2016M_bin8_TTTo2L2Nu
316 prop_binSR_electron2016M_bin3_WpWpJJ_QCD
317 prop_binSR_electron2016M_bin2_WpWpJJ_QCD
318 prop_binSR_electron2016M_bin3_VBS_SSWW_SM
319 prop_binCRWS_electron2016M_bin7_TVX
320 prop_binSR_electron2016M_bin2_TVX
321 prop_binSR_electron2016M_bin3_Triboson
322 prop_binSR_electron2016M_bin2_VBS_SSWW_SM
323 prop_binSR_electron2016M_bin2_Triboson
324 prop_binCRWS_muon2017_bin10_VG
325 prop_binCRWS_electron2016M_bin8_VG
326 prop_binCRWS_muon2017_bin9_TVX
327 prop_binCRWS_electron2016M_bin8_TVX
328 prop_binCRWS_electron2016M_bin9_Triboson
329 prop_binCRWS_muon2016M_bin8_VG
330 prop_binCRWS_muon2016M_bin8_TVX
−2 −1 0 1 2
−0.1 −0.05 0 0.05 0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.7}_{-0.6}$
331 prop_binCRWS_electron2016M_bin8_Triboson
332 prop_binCRWS_electron2016M_bin8_ZZtoLep
333 prop_binCRWS_electron2017_bin9_ZZtoLep
334 prop_binCRWS_muon2017_bin9_Triboson
335 prop_binCRWS_muon2017_bin9_ZZtoLep
336 prop_binCRWS_electron2016M_bin7_Triboson
337 prop_binCRWS_electron2016M_bin7_ZZtoLep
338 prop_binCRWS_electron2016M_bin9_ZZtoLep
339 prop_binCRWS_electron2017_bin9_TVX
340 prop_binCRWS_muon2017_bin10_TVX
341 prop_binCRWS_muon2017_bin10_Triboson
342 FES
343 prop_binCRWS_electron2016M_bin9_TVX
344 prop_binCRWS_electron2017_bin9_Triboson
345 prop_binCRWS_muon2016M_bin8_Triboson
346 prop_binCRWS_electron2016M_bin7_VBS_SSWW_SM
347 prop_binCRWS_electron2016M_bin8_VBS_SSWW_SM
348 prop_binCRWS_electron2017_bin9_VBS_SSWW_SM
349 prop_binCRWS_muon2016M_bin8_VBS_SSWW_SM
350 prop_binCRWS_muon2016M_bin8_ZZtoLep
351 prop_binCRWS_muon2017_bin10_VBS_SSWW_SM
352 prop_binCRWS_muon2017_bin10_ZZtoLep
353 prop_binCRWS_muon2017_bin9_VBS_SSWW_SM
354 QCDScale_WZ
355 pdf_Tot_WZ
356 prop_binCRWS_electron2016M_bin9_VBS_SSWW_SM
357 tau_vsele_2016M
358 tau_vsele_2017
359 tau_vsele_2018
360 tau_vsmu_2016M
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.7}_{-0.6}$
361
tau_vsmu_2017
362
tau_vsmu_2018
−2 −1 0 1 2
−1 −0.5 0 0.5 1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$